

IPE Simulation — Engine Math Reference

Instructor reference (not for students)

This document maps the textbook theory each phase teaches to what the engine actually computes, and flags where the two diverge. It exists so you can answer a sharp question and never overclaim. Students never see it.

1. Modeling philosophy

The engine is a **pedagogical simulator**, not a computable general equilibrium (CGE) or DSGE model. Every round is resolved by tractable closed-form arithmetic — Cobb-Douglas and CES aggregators, multiplicative welfare adjustments, and reduced-form policy-to-state dynamics — chosen so students can *act* and *see* a directional result, deliberately trading equilibrium rigor for legibility.

Two consequences worth holding in mind throughout:

- **Welfare is dimensionless utility**, not money. “Money” quantities (debt, taxes, side payments) reach welfare only through a stated bridge — valuing goods at **world_prices** — or are kept in a separate ledger that never touches utility.
- **Prices are not market-clearing**. Inter-country trade is barter at student-negotiated ratios; there is no endogenous price vector and no factor-price equalization. The engine demonstrates each model’s qualitative lesson; it does not solve for equilibrium.

Notation

g indexes goods (cloth, wine, machinery); j indexes the J goods in welfare; L, K are labor and capital; Y_g is output of good g ; c_j is consumption of good j ; w, r are wage and return to capital; D is a country’s debt stock; d is its currency depreciation factor ($1.00 = \text{par}$).

Parameters (all tunable at the top of engine.py)

Symbol / constant	Value	Meaning
factor shares (cloth)	$\alpha=0.70, \beta=0.30$	labor-intensive
factor shares (wine)	$\alpha=0.55, \beta=0.45$	moderate
factor shares (machinery)	$\alpha=0.25, \beta=0.75$	capital-intensive
WORLD_PRICES	cloth 1, wine 1, machinery 1.5	numeraire goods prices
productivity tiers	HIGH 1.3, MED 1.0, LOW 0.7	firm productivity (Ph. 4)
VARIETY_RHO ρ	0.6	CES; $\sigma = 1/(1 - \rho) = 2.5$
PHASE5_MONEY_GROWTH_CHOICES	0, 2, 5, 10%	money-supply growth
WARNING_DEVALUATION	0.90	stress-1 currency drop
CRISIS_DEVALUATION	0.70	stress-2 currency drop
CRISIS_WELFARE_HIT	0.10	stress-2 welfare hit
BASE_FX_FRICTION	0.02	cross-currency friction
WARNING_FRICTION_BUMP	0.03	friction after a warning
DEBT_BASE_RATE	0.05	base sovereign rate
DEBT_RISK_PREMIUM	0.15	premium $\times$ (debt/capacity)
DEBT_DEFAULT_BAN_ROUNDS	2	borrowing ban after default
DEBT_DEFAULT_FRICTION	0.05	defaulter trade friction
DEBT_DIVIDEND_PENALTY	0.5	WTO dividend halved in ban
IMF_DEBT_RELIEF	0.5	bailout refinances half the stock
IMF_AUSTERITY	0.08	austerity welfare cut
IMF_AUSTERITY_ROUNDS	2	austerity duration
WTO_DIVIDEND	0.02	member-member friction cut
HEGEMON_PROVISION_COST	0.04	hegemon's welfare cost to lead
HEGEMON_PROVISION_BENEFIT	0.05	global friction cut if providing
HEGEMON_WITHHOLD_PENALTY	0.07	global friction rise if withholding

The welfare pipeline (order of operations each round)

For each country the engine computes base utility, then applies multiplicative adjustments in this fixed order (later phases only):

$$U_{\text{base}} \xrightarrow{\text{Ph.5 crisis}} \xrightarrow{\text{Ph.7 hegemon cost}} \xrightarrow{\text{Ph.7 global crisis}} \underbrace{U_{\text{trade}}}_{\text{gains-from-trade measured here}} \xrightarrow{\text{Ph.6 debt}} U_{\text{final}}.$$

Gains-from-trade are measured at U_{trade} — before debt — so that borrowing (consumption pulled from the future) is not miscounted as a gain from trade.

2. Phase 1 — Ricardian comparative advantage

Theory. One factor (labor), constant productivity per good. A country exports the good in which it has lower opportunity cost. The PPF is a straight line; gains from trade come from specialization.

Engine (`_compute_production`, `_utility`).

$$Y_g = L_g \cdot a_g \quad (\text{linear: } a_g = \text{productivity of good } g)$$

$$U = \prod_{j=1}^J c_j^{1/J} \quad (\text{Cobb-Douglas, equal weights})$$

Opportunity cost (shown in the briefs, 2-good case): $OC = a_{g_0}/a_{g_1}$. Trade is barter; a tariff destroys a fraction of the imported good, $\text{received} = \text{qty} (1 - t)$.

Divergence. Linear production $\Rightarrow$ constant opportunity cost (no diminishing returns, no curved PPF). “Gains from trade” is utility of the consumed bundle minus utility of own production (the no-trade baseline), not a welfare-theoretic equivalent variation.

3. Phase 2 — Heckscher-Ohlin

Theory. Two factors. A country exports the good that intensively uses its abundant factor. Stolper-Samuelson: opening trade raises the real return of the abundant factor and lowers the scarce factor (the magnification effect: $\hat{p}_1 \Rightarrow \hat{w} > \hat{p}_1 > 0 > \hat{r}$ when good 1 is labor-intensive).

Engine (`_compute_production`, `_shadow_prices`, `_compute_factor_prices`). Production is Cobb-Douglas; factor returns are *value* marginal products at implicit home prices taken from the post-trade consumption bundle c :

$$\begin{aligned} Y_g &= \text{TFP}_g \cdot L_g^{\alpha_g} K_g^{\beta_g}, & \alpha_g + \beta_g &= 1 \\ p_g &= \min\left(\frac{U}{c_g}, 10\right), & U &= \left(\prod_j c_j\right)^{1/J} \quad (\text{so } \prod_j p_j^{1/J} = 1) \\ w_g &= p_g \alpha_g \frac{Y_g}{L_g}, & r_g &= p_g \beta_g \frac{Y_g}{K_g} \end{aligned}$$

The reported w and r are allocation-weighted averages of w_g and r_g . p_g is the Cobb-Douglas marginal utility $U/(Jc_g)$ rescaled so the ideal price index is 1: a good the country consumes little of is dear at home, and returns are measured in consumption bundles — *real* returns. In autarky $c = Y$, so prices reflect domestic scarcity. Opening trade changes c , hence p , hence w and r : exporting the labor-intensive good raises its home price and with it the wage. *That is the Stolper-Samuelson channel.* The cap of 10 binds only when a good is exported almost entirely.

Why this form. The engine originally reported *physical* marginal products, $w_g = \alpha_g Y_g / L_g$. Those depend only on the allocation, so trade could never move them — in class a country that traded with an unchanged allocation showed identical factor prices to three decimals — and averaging cloth-units with machinery-units is not meaningful. A first fix scaled prices so the bundle was worth its world-price value; nominal returns then swelled and shrank with the gains from trade, and a capital-abundant country could show its return to capital *falling* when trade opened. Normalizing the price index to 1 fixes that: in a controlled test with every allocation held fixed, the labor-abundant countries’ wages rise and capital returns fall, and the capital-abundant country’s capital return rises and wage falls, in absolute real terms (`stress_test.py`, `test_stolper_samuelson`). Histories saved before this change keep the old numbers; replaying those rounds gives the new ones.

Divergence. The prices are shadow prices read off a consumption bundle, not market-clearing prices: there is no zero-profit condition, and because students allocate factors freely, sector marginal products are not equalized — the reported w and r are averages across sectors, not a single economy-wide factor price. No factor-price equalization. The magnification effect holds in the controlled test but is not guaranteed for arbitrary play: when a country reallocates its factors *and* trades in the same round, the reallocation effect and the trade effect are mixed together. To show Stolper-Samuelson cleanly, compare a country whose allocation did not change. Trade is also barter at whatever terms students negotiate, not balanced at common prices, so a lopsided deal is a transfer: w and r value what the country *produces* at home prices, including goods shipped out for little in return, and both can rise while welfare falls. (In class, a country that sent 27 cloth for 9 machinery saw its wage double while its welfare fell 23%.) Welfare, not w and r , says whether a deal paid. The engine will not reproduce, e.g., the Leontief paradox.

4. Phase 3 — MNCs, varieties, new trade theory

Theory. Krugman: increasing returns + love of variety yield intra-industry trade and gains from variety, formalized with CES preferences and monopolistic competition.

Engine (`_compute_firm_production`, `_build_variety_bundles`, `_utility_with_varieties`, `_compute_firm_profits`). MNC output is linear in scale, $Y_{\text{firm}} = \text{scale} \times \text{productivity}$ (0 in a relocation round; scale capped at `max_scale`). Each good's consumption is a CES aggregate over its varieties, nested inside the Cobb-Douglas:

$$c_g = \left(\sum_v q_v^\rho \right)^{1/\rho}, \quad U = \prod_g c_g^{1/J}, \quad \rho = 0.6, \quad \sigma = \frac{1}{1-\rho} = 2.5.$$

More varieties raise c_g at fixed total quantity — the variety gain. Firm profit (numeraire): $\pi = Y_{\text{firm}} \cdot p_{\text{industry}} - \text{scale} \cdot \text{unit_cost}$. Trade moves a good's varieties in proportion to the exporter's variety mix.

Divergence. Real new trade theory has monopolistic competition with free entry, markup pricing, and an *endogenous* number of varieties pinned down by market size and fixed costs. Here the firm set is fixed and CES enters only on the consumption side; firms sell at a fixed world price, not a markup. The model captures love-of-variety and intra-industry trade qualitatively, not the monopolistic-competition equilibrium.

5. Phase 4 — Heterogeneous firms (Melitz)

Theory. Firms draw productivity φ ; only firms above a cutoff φ^* find it worth paying the fixed export cost f_x and exporting. Trade reallocates toward high- φ firms and raises average productivity (selection).

Engine (`_compute_firm_profits`). Each firm has a fixed productivity tier. With `export = True` it pays the fixed cost:

$$\pi = \underbrace{Y_{\text{firm}} p_{\text{ind}}}_{\text{revenue}} - \text{scale} \cdot \text{unit_cost} - \underbrace{f_x}_{\text{if exporting}} - \underbrace{p_{\text{ind}} Y_{\text{firm}} \cdot \tau_{\text{mnc}}}_{\text{MNC tax, Ph.4+}}.$$

Selection is demonstrated by outcome: a low-productivity firm that pays f_x earns negative π ; a high-productivity firm profits. Populist regimes set a tariff floor (applied tariff = $\max(\text{declared}, \text{floor})$) and an MNC revenue tax routed to a **separate ledger** (it does not enter welfare).

Divergence. Real Melitz derives φ^* endogenously from the productivity distribution and fixed costs in equilibrium. Here productivity is discrete (HIGH/MED/LOW) and the export choice is the student's, so selection is shown, not solved.

6. Political approval (all phases)

Theory. In welfare terms a tariff is pure deadweight loss, so a welfare-maximizing player would never protect — which makes the politics of Blocks 4, 6 and 7 unplayable. The missing half is the *distributional* one: protection is a transfer to the import-competing sector, whose gain is concentrated and organized while the consumer's loss is diffuse (Olson). This ledger supplies that political return without touching the utility function.

Engine (`_update_approval`, `_resolve_governments`). For country i , with M_i the value of imports actually received, $\bar{t}_i$ the import-weighted mean tariff levied on them, C_i consumption capacity at

world prices, S_i side payments received, and A_i approval:

$$\begin{aligned}
&\text{exposure } x_i = M_i/C_i, \quad \bar{t}_i = (\text{tariff-weighted}) \\
\Delta A_i = &\underbrace{\pi \bar{t}_i x_i}_{\text{protection}} - \underbrace{\varepsilon x_i}_{\text{openness}} + \underbrace{\text{clip}\left(\rho \frac{U_i - U'_i}{U'_i}\right)}_{\text{prosperity}} + \underbrace{\kappa S_i/C_i}_{\text{compensation}} + \underbrace{\delta (A_0 - A_i)}_{\text{drift}} \\
A_i \leftarrow &\text{clip}(A_i + \Delta A_i, 0, 100)
\end{aligned}$$

with $\pi = 35$, $\varepsilon = 14$, $\rho = 35$ (clipped to ± 15), $\kappa = 25$, $\delta = 0.10$, $A_0 = 50$, and U'_i last round's welfare.

The endogenous backlash. If $A_i < 30$ for two consecutive rounds the government falls: `inject_populist_backlash` fires automatically (20% tariff floor, 10% MNC tax) and A_i resets to A_0 . Protection is therefore imposed *on* a country that ignores its losers — the instructor does not choose the victim.

Tuning. Holding policy fixed, approval converges to

$$A^* = A_0 - \frac{x(\varepsilon - \pi \bar{t})}{\delta},$$

so the steady state, not the starting value, decides who survives. The coefficients are set so A^* sits below the crisis floor under free trade (≈ 20) and above it from roughly a 20% tariff (≈ 35). Under uniform play over a 20-round term this yields: free trade highest welfare but one government falls; 20% tariffs all survive at $\sim 3\%$ less welfare; 40% tariffs politically safe at $\sim 12\%$ less. *An earlier tuning put A^* below the floor at every tariff level, so every trading country fell regardless of policy — a doom clock rather than a decision. Check A^* , not the first few rounds, after changing any coefficient.*

Divergence. This is the most openly *reduced-form* component in the engine — a stipulated response function, in the same family as trilemma stress and the debt risk premium, not something derived from preferences or voting. There is no median voter, no electoral rule, and no separate interest groups: the “concentrated benefit” is a coefficient, not an aggregation over agents. It also has no free-rider problem and no lobbying cost, so it dramatizes Olson’s logic rather than modelling it. Approval never enters U , so all welfare and gains-from-trade figures remain exactly as documented above; it is a second scoreboard, deliberately in tension with the first.

7. Phase 5 — Money, FX, and the trilemma

Theory. The Mundell-Fleming “impossible trinity”: a country cannot simultaneously run a fixed exchange rate, open capital markets, and independent monetary policy. Pushing all three invites a speculative attack.

Engine (`_apply_monetary_decisions`, `_compute_fx_friction`). Let s be the stress counter and d the depreciation factor. Overreach is $\text{peg} \wedge \text{open-capital} \wedge \text{independent-money}$. Per round:

$$\text{overreach} \Rightarrow s += 1; \quad \text{else } s = 0.$$

$$s = 1 : d \times = 0.90 \text{ (warning);} \quad s = 2 : d \times = 0.70, U \times = 0.90, s \rightarrow 0 \text{ (crisis).}$$

Money-supply decay: if independent, $d \times = (1 - g)$ with $g \in \{0, 2, 5, 10\%\}$. FX friction on a trade leg (`_compute_fx_friction`): 0 if the reserve-currency holder is a party or the two share a monetary

union; otherwise `BASE_FX_FRICTION` = 0.02, halved if both peg, plus `WARNING_FRICTION_BUMP` = 0.03 if either was just warned. Friction stacks on tariffs multiplicatively: $\text{loss} = 1 - (1 - t)(1 - \text{fx})$.

Divergence. d is a *reduced-form currency-strength index* driven by policy choices and shocks — there is no money supply/demand, no interest parity, no PPP, no modeled exchange-rate market. The trilemma is operationalized as a stress accumulator, not derived from an open-economy equilibrium. A monetary union shares one (s, d, policy) state across members (individual debts, shared currency — the Eurozone analogy).

8. Phase 6 — Sovereign debt

Theory. Debt is sustainable while the interest burden stays within capacity; spreads rise with the debt burden; “original sin” (Eichengreen-Hausmann) means foreign-currency debt becomes crushing when the local currency depreciates.

Engine (`_apply_country_debt`). Let consumption capacity $C = \sum_g c_g p_g$ (goods at world prices). Then:

$$\begin{aligned} \text{rate} &= \underbrace{0.05}_{\text{base}} + \underbrace{0.15}_{\text{premium}} \cdot \frac{D}{C}, & \text{interest} &= D \cdot \text{rate} \\ \text{borrow } B : & U \times = \left(1 + \frac{B}{C}\right), & B &\leq C \\ \text{service (interest + repay)} : & U \times = \max\left(0, 1 - \frac{\text{service}/d}{C}\right), & D &\leftarrow D + B - \text{repay} \end{aligned}$$

The division of service by d is original sin: a weak currency ($d < 1$) inflates the real burden of hard-currency debt. **Default** wipes the stock ($D \rightarrow 0$), bans borrowing for 2 rounds, and adds trade friction, but skips the service hit that round. **IMF bailout**: $D \times = 0.5$ now, in exchange for $U \times = 0.92$ for 2 rounds (austerity).

Divergence. No inter-temporal optimization (no Euler equation, no forward-looking default decision); the borrow/service welfare scaling is a mechanical multiplier. Debt is denominated only in the reserve currency. The rate rule is a reduced-form spread, not a default-probability-priced yield.

9. Phase 7 — Institutions and power

Theory. Hegemonic stability (Kindleberger, Krasner): a leader supplies public goods — open markets, a stable system — at private cost, and is tempted to free-ride. Regimes/the WTO (Keohane) lower transaction costs among members.

Engine (`_institutional_friction_delta`, `_global_crisis_factor`, `challenge_hegemon`). All institutional effects act on *trade friction* or welfare. The friction added to a trade leg, on top of the Phase-5 base and any Phase-6 default friction, is

$$\Delta_{\text{inst}} = \underbrace{(-0.05 \text{ if provide else } +0.07)}_{\text{hegemon public good}} + \underbrace{-0.02}_{\substack{\text{WTO dividend, if both members} \\ \& \text{neither defected} \\ (\times 0.5 \text{ if debt-banned})}},$$

then the total fx is clamped to $[0, 1]$. The hegemon pays $U \times = 0.96$ when it provides. A pending **global crisis** of severity ς scales every country’s welfare by

$$1 - \varsigma \cdot (0.5 \text{ if hegemon provides}) \cdot (1 - 0.5 \text{ wto_share}),$$

so leadership and broad membership both soften it. A **hegemonic challenge** succeeds when the challenger coalition’s cumulative welfare exceeds that of all non-coalition countries; hegemony and the reserve-currency role then transfer. Side payments are goods transfers applied before welfare.

Divergence. A stylized public-goods/coordination model, not a game-theoretic equilibrium. The challenge rule is a raw weight comparison, not a bargaining solution; “defection” is simply applying a tariff above a binding. The hegemony parameters are tuned (see the simulation notes) so that provision is collectively beneficial in both WTO-none and WTO-all settings while the hegemon is still individually tempted to free-ride.

All formulas above are exactly what the code computes; the `stress_test_phase*.py` suites assert these numbers. To change a behavior, edit the constant at the top of `engine.py` and rerun the tests.